Data Storm v7.0

- Preliminary Round –

Organized By

Rotaract Club of University of Moratuwa

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1. Background

1.1. Business Context

A leading beverage manufacturer in Sri Lanka operates a massive distribution network spanning over 80,000 traditional retail outlets from bustling urban grocery stores in Colombo to small “kades” (corner shops) and local eateries in rural outstations.

Currently, sales teams allocate trade marketing budgets, coolers, and promotional discounts based on historical sales averages. However, leadership realizes this is fundamentally flawed. Historical sales only reflect what an outlet did sell, not what it could sell. A high-traffic town-center kade might be underperforming due to poor stock management or credit constraints, while a small village shop might already be maxed out.

The company wants to shift from historical-based resource allocation to Potential-Based Allocation. They need to predict the Maximum Monthly Purchase Potential of every traditional trade outlet to optimize trade spend and cooler deployments.

1.2. Current Scope

For this challenge, we are focusing on **20,000 traditional trade outlets** (kades, groceries, eateries, pharmacies, etc.) serviced by **10 key distributors** across 4 key provinces in Sri Lanka.

Province	# Distributors	Distributor IDs
Western	3	'DIST_W_01', 'DIST_W_02', 'DIST_W_03'
Central	3	'DIST_C_01', 'DIST_C_02', 'DIST_C_03'
North-Western	2	'DIST_NW_01', 'DIST_NW_02'
Southern	2	'DIST_S_01', 'DIST_S_02'

2. Preliminary Problem Statement

Your objective is to design a robust analytical framework and model to estimate the latent maximum monthly volume potential (in liters) for our retail network for the month of **January 2026**.

Here is the reality of this challenge: There is no perfect “ground truth,” and we do not have a secret answer key. “Potential” is a theoretical ceiling.

“Potential” is a hidden variable. You are not given a target variable (y). The historical volume is censored; it represents the minimum of either:

- **(A)** The true consumer demand.
- **(B)** Systemic Constraints (credit limits, stockouts, or delivery caps).

Your goal is to build the most logical, defensible, and mathematically sound framework to uncap this latent demand.

3. Data Description

3.1. Dataset Description

Participants are provided with the following datasets:

1. **transactions_history_final.csv**: Granular outlet-level data.
2. **outlet_master.csv**: Outlet related data.
3. **outlet_master.csv**: Longitude and Latitude for each outlet.
4. **distributor_seasonality_details.csv**: Month specific seasonality for distributors.
5. **holiday_list.csv:** List of holidays

3.2. The External Data Challenge (POI Scraping)

We are **not** providing Point of Interest (POI) data. Teams can use public APIs (like OpenStreetMap, Overpass) or web scraping to identify nearby schools, bus stands, hospitals, tourist attractions, etc. which will help with getting the potential.

3.3. The Data Quality Reality

Please note that all the datasets **are unprocessed, raw exports** from legacy Sales Force Automation (SFA) and distributor ERP systems. It has not been sanitized.

In the FMCG sector, system data is notoriously messy. It is subject to connectivity blackouts, human data-entry shortcuts, automated ghost entries, and master-data decay. **Do not trust the data blindly**. Your first major task is data forensics: you must separate true market signals from system artifacts and human errors before building your potential framework.

4. Technical and Architecture Requirements

In addition to the analytical challenge, participants must demonstrate sound Data Engineering practices and meaningful AI integration. The following are requirements for all submissions.

4.1. Lakehouse Pipeline Architecture

Participants should structure their codebase into three clearly defined layers, mirroring industry-standard data Lakehouse practices.

- **Bronze (Raw Ingestion):** Ingest all flat files as-is with no transformations. This layer preserves the original data exactly as provided.
- **Silver (Cleaned):** Apply all DE checks and data cleaning logic to produce sanitized datasets. Records failing checks must be quarantined into a separate rejected records store with a documented failure reason they must not be silently dropped or carried forward.
- **Gold (Enriched):** Apply feature engineering and produce model-ready output, drawing from all Silver-layer datasets including both internally provided and externally scraped data.

Note: A cleanly structured local directory setup (e.g., storing intermediate .csv or. parquet files in dedicated folders) is perfectly sufficient.

4.2. Reusable Data Quality Checks

Participants must implement data quality checks as reusable, parameterizable functions applied consistently across all datasets. The following are examples of checks to consider

This is not a comprehensive list, and teams are encouraged to identify and implement additional checks relevant to the data:

- **Duplicate Check:** Detect duplicate records based on a configurable primary key (single or composite).

- **Null Check:** Flag records where mandatory fields contain null or empty values.
- **Referential Integrity Check:** Validate that foreign key values in one dataset exist in a reference dataset.
- **Value Range Check:** Assert that numeric fields fall within an expected min/max boundary.
- **Format / Type Check:** Validate that fields conform to an expected data type or format (e.g. dates, IDs).

5. Deliverables

This challenge mimics the reality of a modern data science, data engineer role, there is no hidden “answer key” to blindly optimize against. Therefore, evaluation is 100% qualitative, focusing strictly on your data forensics methodology, data Lakehouse implementation, code and analytical robustness, and business logic.

To be considered for the judging round, teams must submit the following three deliverables:

1. **The “Latent Potential” Output (CSV):** A single file ***(teamname_predictions.csv)*** containing your final, uncapped volume predictions. It must include Outlet_ID and your predicted Maximum_Monthly_Liters (Potential) for the month of **January 2026**.

2. **The Reproducible Codebase (GitHub Link or Zipped Repo):** Your complete code (Python scripts / Jupyter Notebooks, etc.) used for data cleaning, POI scraping, and modeling. Must include:
   a. **A README.md** with clear instructions on how to run your pipeline end to end.
   b. A clearly structured codebase reflecting the Bronze → Silver → Gold layer separation.
   c. A clearly structured modeling/analytical pipelines from simple EDAs to final prediction.

3. **PDF Summary Report:** A comprehensive technical document (**max 5 pages**, including the cover page) This document must explicitly address the following:
   a. **Data Forensics and Hygiene:** Document the specific system anomalies trapped, how records were quarantined, and your DE checks setup.
   b. **POI Data Acquisition:** Detail your technical approach to acquiring external geospatial signals (APIs, scraping methods or tools) and document type of POIs you targeted as catchment drivers and how you mapped them to the internal outlets.

c. **Causal Base Logic:** Detail your initial statistical or probabilistic approach to solving the left-censored demand curve and explain the exact methodology used to calculate the uncapped potential ceiling.
d. **GenAI Transparency Log:** Provide a dedicated log outlining how, where, and why your team utilized Generative AI during the 36 hours. Document on how LLMs were used as engineering accelerators.

6. Evaluation Metrics

The judging panel of data scientists, data engineers and business leaders will evaluate teams on following criteria:

Criteria	Weight
Data Engineering and Forensics (Cleaning)	40%
Methodology and Base Math	40%
Generative AI Utilization and Workflow	20%

1. Data Engineering and Forensics (40%)

- Did the team implement a clearly structured Bronze → Silver → Gold pipeline with a rejected records store?
- Was the DE checks implementation reusable, parameterizable, and applied consistently across datasets?
- Did the team successfully identify, clean and neutralize legacy system artifacts?
- How robust was the web-scraping / API pipeline for external POI data?
- Did the team engineer create highly relevant features to isolate true market signals?

2. Methodology and Base Math (40%)

- How did the team conceptualize the problem of "Latent Potential"?
- What mathematical or statistical approach was used to handle the missing target variable and right-censored data?

4. Generative AI Utilization and Workflow (20%)

- Did the team clearly document how, where, and why they used LLMs (e.g., Gemini, Copilot, ChatGPT) during the hackathon?

- Was AI used intelligently as an accelerator (e.g., for brainstorming causal frameworks, writing boilerplate scraping code, or debugging complex statistical logic) rather than a crutch?
- Is there evidence that the team critically evaluated, iteratively prompted, and rigorously validated the AI-generated code and assumptions, rather than blindly trusting the output?

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